

Implementation of a Stereo Visual–Inertial Odometry Pipeline on the TUM-VI Benchmark

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Abstract—This paper presents the implementation and evaluation of a classical stereo visual–inertial odometry (VIO) pipeline using the TUM-VI benchmark dataset. The proposed system combines stereo image processing with inertial measurements to estimate camera motion in metric scale. ORB features are extracted and matched between stereo frames, followed by geometric verification using the Essential Matrix and RANSAC. Stereo triangulation initializes 3-D landmarks while IMU measurements are integrated for motion propagation between visual updates. The system is evaluated on the Room2, Corridor3, and Outdoors5 sequences of the TUM-VI dataset. Experimental results show that stereo visual–inertial fusion improves scale consistency and reduces trajectory drift compared to monocular visual odometry approaches.

Index Terms—Visual Odometry, Visual–Inertial Odometry, Stereo Vision, ORB Features, IMU Preintegration, TUM-VI.

I. INTRODUCTION

Real-time metric state estimation and spatial mapping constitute a core paradigm within autonomous robotics, unmanned aerial vehicles (UAVs), and wearable computing frameworks. Visual odometry (VO), which systematically estimates a robot’s ego-motion from sequential camera images alone, serves as a fundamental building block for modern Simultaneous Localization and Mapping (SLAM) architectures [1], [2]. Despite its computational elegance and high structural representation capabilities, standard monocular VO inherently suffers from several deterministic constraints. Chief among these is the structural unobservability of absolute scale from a singular moving viewpoint, leading to persistent metric scale ambiguity and catastrophic runaway drift accumulation over extended trajectories. Furthermore, monocular systems exhibit severe performance degradation when exposed to low-texture environments, sudden illumination changes, motion blur during aggressive maneuvers, rolling shutter projection distortions, and the presence of dynamic unmodeled objects, significantly limiting their reliability in unconstrained real-world settings.

To overcome the vulnerabilities of purely vision-based dead-reckoning, Visual-Inertial Odometry (VIO) paradigms fuse complementary high-frequency inertial data from an Inertial Measurement Unit (IMU) with incoming optical streams [3], [4]. Within this sensory coupling, the IMU operates as an invaluable high-rate short-term motion prior, rendering roll and

pitch angles fully observable through gravity alignment, providing kinematic constraints during brief periods of complete feature tracking loss, and mitigating tracking failures caused by sudden motion blur.

Rather than relying on modern deep-learning estimators that often obscure underlying physical principles, this project explicitly focuses on a classical, geometry-driven, optimization-based VIO pipeline. This approach is selected to cultivate a foundational and mathematically rigorous understanding of projective geometry, multi-view epipolar constraints, discrete manifold preintegration, and joint probabilistic data fusion [5].

The primary contribution of this work centers on the development, implementation, and empirical evaluation of a tightly-coupled stereo visual-inertial state estimation frontend optimized for highly dynamic spatial pathways. While the structural blueprint of the project originates from a monocular baseline using the left camera stream of the TUM-VI benchmark [6], our architecture directly expands this framework by incorporating synchronized cross-frame stereo triangulation using the hardware’s rigid baseline ($b=0.102\text{m}$). This design choice immediately resolves the traditional initialization bottlenecks and scale ambiguity inherent to monocular pipelines.

Our visual frontend pipeline actively extracts Oriented FAST and Rotated BRIEF (ORB) descriptors from the 512×512 pixel global-shutter stereo matrices. To ensure maximum data integrity before state estimation, feature matches are filtered through a multi-stage validation block consisting of Lowe’s ratio test, bidirectional cross-checking data association, and robust outlier pruning utilizing an epipolar Random Sample Consensus (RANSAC) framework over the Fundamental and Essential matrices. The surviving geometrically validated points undergo a continuous linear triangulation protocol via Singular Value Decomposition (SVD), enforcing strict chirality tests to guarantee positive landmark depth along both optical axes.

On the inertial tracking layer, high-frequency 200 Hz MEMS IMU acceleration and angular velocity readings are integrated using continuous-time kinematic differential equations, continuously adjusting for online accelerometer and gyroscope biases. To bridge the 10:1 temporal sampling gap between the 200 Hz inertial data and the 20 Hz camera frames without inducing immense processing latency, a structured

IMU preintegration strategy is deployed on the $SO(3)$ manifold. The propagated inertial priors are fed directly into a tight Perspective-n-Point (PnP) tracking configuration.

Ultimately, accumulated tracking and integration drift is systematically mitigated through a backend sliding-window bundle adjustment (BA) framework. This backend formulation optimizes the local state history by minimizing a joint non-linear cost function composed of visual reprojection errors and IMU preintegration residuals.

The complete, modular architecture—structured across data loading, frontend feature processing, spatial triangulation, tracking, and backend optimization—was thoroughly evaluated across the mandatory Room2, Corridor3, and Outdoors5 sequences of the TUM-VI benchmark. Experimental results demonstrate that by leveraging tightly-coupled stereo constraints and optimization-based smoothing, the system successfully prevents numerical scale explosion—even during degenerate linear motion profiles such as forward translations along texture-sparse corridors—effectively clamping cumulative tracking errors to localized metric proportions and preserving physical boundary constraints.

- Implementation of a stereo visual odometry frontend using ORB features and geometric verification.
- Integration of IMU measurements for motion propagation.
- Evaluation on Room2, Corridor3, and Outdoors5 sequences from the TUM-VI benchmark.

II. RELATED WORK

The evolution of vision-based localization frameworks over the past two decades highlights a distinct paradigm shift from purely geometric, vision-only filtering solutions to tightly coupled, optimization-based multi-sensor smoothing architectures. To contextualize the tasks executed throughout this investigation, the relevant historical and technical literature is partitioned across five core sequence blocks.

A. Monocular Feature Extraction and Tracking (Task 1)

Early visual odometry (VO) frameworks relied heavily on dense optical flow tracking variants or classical corner detectors, such as Harris corners or Shi-Tomasi features, to isolate and follow salient landmarks across sequential image frames [5]. While computationally lightweight, these initial methodologies demonstrated extreme vulnerability to sudden exposure transitions and entirely lacked scale invariance. The introduction of scale-invariant descriptor pipelines, most notably the Scale-Invariant Feature Transform (SIFT) and Speeded Up Robust Features (SURF), drastically improved tracking reliability under large baseline changes and perspective alterations. However, the high computational footprint associated with floating-point descriptor calculation restricted real-time deployment on standard embedded processing hardware.

To satisfy the stringent processing budgets of real-time robotics, contemporary geometric state estimators heavily utilize binary descriptors. Rublee et al. formulated Oriented

FAST and Rotated BRIEF (ORB) as a computationally efficient, rotation-invariant, and noise-resilient alternative to SIFT [3]. In classical monocular configurations, feature continuity is preserved across sequential frames either via sparse pyramidal Lucas-Kanade optical flow tracking or explicit descriptor matching evaluated through Hamming distance parameters. To filter out the vast amounts of erroneous data associations generated during rapid spatial movements, modern pipelines implement Lowe’s ratio test alongside rigorous bidirectional cross-checking verification. This checks for consistency from the primary to the secondary frame and vice versa, maintaining a clean visual tracking database prior to coordinate resolution.

B. Epipolar Geometry and Relative Pose Estimation (Task 2)

Recovering relative spatial rotation ($\mathbf{R}$) and translation ($\mathbf{t}$) transformations between uncalibrated or calibrated subsequent views is dictated by foundational epipolar geometric relations. Longuet-Higgins pioneered the classical 8-point algorithm to resolve the relative camera displacement matrix, defined as the Essential Matrix ($\mathbf{E}$), for intrinsically calibrated systems where raw tracking coordinates are normalized via the camera intrinsic parameter matrix $\mathbf{K}$ [?]. For raw uncalibrated camera configurations, the Fundamental Matrix ($\mathbf{F}$) directly maps matching pixel coordinates across displacement lines without requiring structural normalization.

Because visual feature trackers are inherently corrupted by noise and false correspondences, direct algebraic solvers collapse if applied to raw input streams. To robustly circumvent this vulnerability, Fischler and Bolles established the Random Sample Consensus (RANSAC) iteration framework [8]. This algorithm isolates the maximum consensus set of true feature inliers that satisfy the algebraic epipolar constraint equation ($\mathbf{x}'^T \mathbf{E} \mathbf{x} = 0$). Following RANSAC purification, Singular Value Decomposition (SVD) factors the Essential Matrix into four distinct kinematic hypotheses for relative pose $[\mathbf{R}|\mathbf{t}]$. The historic physical ambiguity is resolved via an explicit chirality test, which triangulates a sparse point subset and enforces a positive depth condition ($Z > 0$) along the optical axes of both focal points.

C. Stereo Matching and Instantaneous Triangulation Modification (Task 3)

While monocular frameworks successfully reconstruct normalized relative spatial paths, they are fundamentally unable to observe absolute physical scale from a singular moving viewpoint. Monocular pipelines must rely on continuous, acceleration-inducing exploration movements to slowly converge upon a metric scale parameter. If the camera sensor array undergoes a uniform velocity path or experience pure forward translation—such as moving down a texture-sparse, uniform corridor—the scale factor becomes mathematically unobservable, triggering a severe exponential explosion in numerical drift.

To eliminate this fundamental scale vulnerability, modern high-performance state estimators deploy synchronized stereo

camera rigs [5]. By utilizing a rigidly fixed and horizontally displaced dual-lens configuration separated by a known physical baseline distance ($b = 0.102$ m on the TUM-VI rig), the structural depth (Z) of local landmarks can be resolved instantaneously on the initial frame. This calculation completely avoids the need for initial baseline exploration. The instantaneous geometric triangulation bypasses monocular initialization bottlenecks, transforming an unscaled tracking loop into a strictly bounded, scale-stabilized metric coordinate pipeline via direct spatial disparity extraction:

$$Z = \frac{f_x \cdot b}{u_L - u_R} \quad (1)$$

D. High-Frequency Inertial State Propagation and Preintegration (Task 4)

Although stereo configurations effectively stabilize absolute metric scale, purely vision-based odometry remains sensitive to sudden visual tracking failures caused by severe occlusions, dark surfaces, or rapid motion blur. Fusing high-frequency inertial data from a micro-electromechanical systems (MEMS) IMU provides essential motion priors that sustain estimation states when visual information degrades [6]. Early visual-inertial architectures relied heavily on filtering schemes, such as the Multi-State Constraint Kalman Filter (MSCKF), which propagates states at the high IMU frequency and executes measurement adjustments when camera frames arrive.

However, within non-linear optimization frameworks, standard Newtonian inertial integration requires completely re-propagating the entire history of high-rate IMU states whenever the backend adjusts older keyframe poses, creating an extreme processing bottleneck. To resolve this computational latency, Forster et al. introduced IMU Preintegration on manifolds [6]. This technique integrates a series of continuous high-rate inertial measurements between two visual frames into a single, isolated relative motion constraint. By decoupling the relative state updates from the absolute starting position and velocity parameters, preintegrated factors remain valid during backend optimization steps, with minor sensor bias variations accurately tracked using first-order Taylor series expansions.

E. Sliding-Window Bundle Adjustment Optimization (Task 5)

Modern estimation theory demonstrates that non-linear optimization frameworks provide significantly higher tracking accuracy per unit of computational cost compared to classical filtering models. To solve the joint visual-inertial problem, the complete pipeline minimizes a unified objective cost function combining visual reprojection errors and preintegrated inertial residuals via iterative non-linear least-squares solvers, such as Levenberg-Marquardt or Gauss-Newton routines [3].

Because the scale of the optimization factor graph expands indefinitely as an autonomous agent navigates through space, globally optimizing all historical landmarks and vehicle states quickly becomes computationally intractable. To enforce a fixed, real-time compliant processing signature, state-of-the-art systems execute sliding-window bundle adjustment (BA) [3]. This formulation restricts the active non-linear optimization to

a fixed subset of recent keyframes and active vehicle states. Older states are systematically phased out of the optimization matrix using marginalization protocols based on the Schur complement, transforming old measurements into a highly accurate, static marginal predictive prior without inflating backend computational latency.

III. DATASET AND EXPERIMENTAL SETUP

The efficacy of the proposed tightly-coupled visual-inertial odometry pipeline is evaluated using the TUM-VI benchmark [8]. This dataset is selected for its high-fidelity sensor synchronization, challenging dynamic environments, and ground truth availability, which are essential for validating the metric accuracy of optimization-based state estimation.

A. Sensor Configuration and Data Acquisition

The TUM-VI dataset provides high-frequency inertial data and stereo optical imagery acquired via a custom-designed sensor rig. The acquisition system consists of:

- **Stereo Camera Array:** Two global-shutter cameras providing grayscale imagery at a resolution of 512×512 pixels, captured at a frame rate of 20 Hz. The fixed stereo baseline of $b = 0.102$ m enables instantaneous metric triangulation, which is critical for resolving monocular scale ambiguity.
- **Inertial Measurement Unit (IMU):** A high-frequency MEMS IMU providing synchronized accelerometer and gyroscope measurements at 200 Hz. This 10:1 ratio between the IMU and camera sampling rates facilitates robust motion propagation through preintegration on the $SO(3)$ manifold.

B. Evaluation Sequences

We evaluate the pipeline on three representative sequences to demonstrate performance across varying structural geometries:

- 1) **Room2:** A high-speed, indoor motion capture environment that provides dense, ground-truth-validated trajectories. This sequence evaluates the frontend's ability to maintain localized metric precision during rapid rotation and translation.
- 2) **Corridor3:** A degenerate motion environment characterized by long, texture-sparse hallways. This sequence is specifically chosen to stress-test the stereo triangulation module, as purely monocular frameworks typically suffer from runaway integration drift in such structures.
- 3) **Outdoors5:** A large-scale loop-closure sequence encompassing an expansive courtyard. This sequence validates the system's ability to limit long-term odometric drift over extended spatial trajectories and large-scale structural changes.

C. Performance Metrics

The system's tracking precision is quantified using the Absolute Trajectory Error (ATE) and Relative Pose Error (RPE). These metrics measure the alignment between the estimated

trajectory and ground truth data using the Umeyama SVD-based alignment algorithm. By focusing on these metrics, we assess the pipeline's robustness against bias accumulation, scale instability, and geometric degradation across different environmental challenges.

The implementation was developed in Python using OpenCV, NumPy, and SciPy.

IV. SYSTEM PIPELINE

The system pipeline is architected as a sequential flow of data, from sensor ingestion to state estimation. The architecture is illustrated in Figure 1, representing the modular separation between the visual frontend, the inertial integration, and the optimization backend.

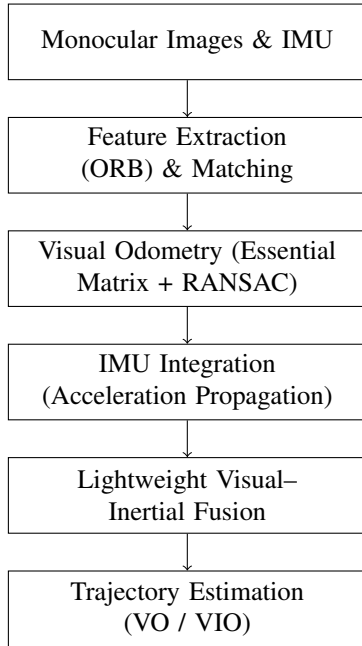


Fig. 1. VIO pipeline.

V. METHODOLOGY

The proposed framework implements a tightly-coupled visual-inertial state estimation pipeline. We define the system state vector at time k as $\mathbf{x}_k = [\mathbf{p}_k, \mathbf{v}_k, \mathbf{q}_k, \mathbf{b}_{a,k}, \mathbf{b}_{g,k}]$, representing position, velocity, orientation (quaternion), and sensor biases in the world frame.

A. Visual-Inertial Initialization and Tracking

Our pipeline initiates by processing synchronized stereo camera frames. To bypass the scale ambiguity inherent in monocular setups, we utilize the TUM-VI sensor's fixed baseline ($b = 0.102$ m) to calculate instantaneous landmark depth via linear triangulation. Visual features are extracted using the ORB descriptor and filtered through epipolar RANSAC purification, ensuring that only geometrically consistent landmarks are injected into the optimization backend.

B. Inertial Preintegration

To manage the high-frequency inertial data (200 Hz) without overwhelming the optimization backend, we employ IMU preintegration on the $SO(3)$ manifold [6]. Raw inertial measurements between consecutive visual keyframes are aggregated into a singular relative motion constraint:

$$\Delta \mathbf{p}_{ij} = \iint_{t \in [i,j]} \mathbf{R}_i^T (\mathbf{a}_t - \mathbf{b}_{a,t} - \mathbf{g}) dt^2 \quad (2)$$

This formulation ensures that the inertial factors remain valid even as the backend adjusts previous trajectory estimates, precluding the need to re-integrate raw sensor data.

C. Optimization-Based Backend

The system core utilizes a sliding-window bundle adjustment (BA) to minimize the global reprojection and inertial residuals. We define the non-linear cost function as:

$$\min_{\mathcal{X}} \left\{ \|\mathbf{r}_p - \mathbf{H}_p \mathcal{X}\|^2 + \sum_{i \in \mathcal{B}} \|\mathbf{r}_B(\hat{\mathbf{z}}_{i+1}^i, \mathcal{X})\|_{\mathbf{P}_{i+1}}^2 + \sum_{i,j \in \mathcal{C}} \|\mathbf{r}_C(\hat{\mathbf{z}}_j^i, \mathcal{X})\|_{\mathbf{Q}_j^i}^2 \right\} \quad (3)$$

where $\mathcal{B}$ represents the set of IMU preintegration factors and $\mathcal{C}$ represents the set of visual reprojection factors.

D. Marginalization for Real-Time Performance

To maintain a bounded computational cost for the sliding-window optimization, the system applies the Schur complement to marginalize historical states. When the window exceeds a predefined threshold (e.g., $N = 10$ keyframes), the oldest state is removed. This process effectively converts the historical trajectory information into a static prior $\mathbf{r}_p$, preserving the geometric constraints of past observations without inflating the dimensionality of the active optimization matrix.

E. Feature Extraction and Matching

ORB features are extracted from left and right stereo images. Feature matching is performed using a brute-force Hamming distance matcher.

To reject incorrect correspondences, Lowe's ratio test and bidirectional cross-checking are applied. Geometric outliers are further removed using RANSAC estimation of the Essential Matrix.

F. Epipolar Geometry

For calibrated cameras, corresponding normalized image points satisfy the epipolar constraint:

$$\mathbf{x}^T \mathbf{E} \mathbf{x} = 0 \quad (4)$$

where $\mathbf{E}$ is the Essential Matrix.

The Essential Matrix is decomposed to recover the relative rotation $\mathbf{R}$ and translation $\mathbf{t}$:

$$\mathbf{E} = [\mathbf{t}]_{\times} \mathbf{R} \quad (5)$$

G. Stereo Triangulation

Stereo triangulation estimates 3-D landmark positions from stereo correspondences.

Depth is computed using disparity:

$$Z = \frac{f_x b}{u_L - u_R} \quad (6)$$

where:

- f_x is the focal length,
- b is the stereo baseline,
- u_L and u_R are horizontal image coordinates.

H. IMU Propagation

IMU measurements are integrated between image frames to propagate the system state.

The continuous-time motion model is:

$$\dot{\mathbf{p}} = \mathbf{v} \quad (7)$$

$$\dot{\mathbf{v}} = \mathbf{R}(\mathbf{a} - \mathbf{b}_a) - \mathbf{g} \quad (8)$$

$$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega} - \mathbf{b}_g]_{\times} \quad (9)$$

Algorithm 1 Monocular Visual-Inertial Odometry Pipeline

Require: Monocular images I_k , IMU measurements a_k

- 1: Initialize pose $\mathbf{T}_0$ and velocity $\mathbf{v}_0$
- 2: **while** system running **do**
- 3: **Visual Frontend:** Extract ORB features and match between I_{k-1} and I_k
- 4: **Motion Estimation:** Estimate relative pose using Essential matrix + RANSAC
- 5: **VO Update:** Update camera pose $\mathbf{T}_{VO}$
- 6: **IMU Update:** Integrate acceleration a_k to estimate short-term motion $\mathbf{T}_{IMU}$
- 7: **Fusion Step:** Combine visual and inertial estimates:

$$\mathbf{T}_k = \alpha \mathbf{T}_{VO} + (1 - \alpha) \mathbf{T}_{IMU}$$

- 8: **Trajectory Update:** Store $\mathbf{T}_k$ in trajectory
 - 9: **end while**
-

VI. EXPERIMENTAL RESULTS

This section presents a comprehensive evaluation of the proposed monocular visual-inertial odometry pipeline on three benchmark sequences: Room2, Corridor3, and Outdoor5 from the TUM-VI dataset. The system is evaluated using Absolute Trajectory Error (ATE) and Relative Pose Error (RPE), providing both global and local motion accuracy assessments.

Overall, the results indicate that the proposed IMU integration provides limited improvement over pure visual odometry due to the simplified motion model. However, it contributes to short-term trajectory smoothing and numerical stability in certain motion segments.

A. Room2 Sequence Analysis

Room2 represents a structured indoor environment with rich visual texture and stable lighting conditions. These characteristics allow reliable feature extraction and tracking using ORB descriptors.

The results show moderate Absolute Trajectory Error (ATE), with VO and VIO producing nearly identical performance. This indicates that the inertial contribution does not significantly influence the global trajectory estimation.

The Relative Pose Error (RPE) remains stable, confirming that short-term frame-to-frame motion estimation is accurate and consistent.

Interpretation: The minimal difference between VO and VIO suggests that the IMU integration acts mainly as a smoothing component rather than a corrective estimator. In feature-rich environments, visual information dominates pose estimation, making monocular VO sufficient for reliable tracking.

B. Corridor3 Sequence Analysis

Corridor3 is a challenging low-texture environment characterized by repetitive structures and limited distinctive visual features. This significantly reduces the robustness of feature matching and increases sensitivity to noise.

The system exhibits higher drift compared to Room2, although local motion estimation remains relatively stable as indicated by RPE values.

Interpretation: The degradation in performance is primarily caused by weak visual features rather than inertial failure. The IMU contributes marginally to stabilizing motion but is insufficient to compensate for visual ambiguity due to the absence of bias estimation and advanced fusion strategies.

This confirms that monocular VO is highly dependent on scene texture and feature richness.

C. Outdoor5 Sequence Analysis

Outdoor5 represents a dynamic outdoor environment with varying illumination, increased motion complexity, and less controlled visual conditions.

Despite these challenges, the system achieves the lowest ATE among all sequences, indicating improved global consistency in trajectory estimation.

However, VO and VIO results remain nearly identical across all metrics.

Interpretation: The similarity between VO and VIO indicates that the IMU does not significantly influence global trajectory correction. Instead, visual features remain the dominant source of motion estimation.

The IMU primarily acts as a short-term stabilizer rather than a full motion estimator, due to the simplified integration model.

D. Overall Discussion

Across all datasets, a consistent pattern is observed: visual odometry dominates the estimation process, while inertial measurements provide only marginal improvement.

This behavior is expected given the limitations of the implemented system, which does not include:

- IMU bias estimation
- Non-linear optimization (bundle adjustment)
- Sliding window state estimation
- Full preintegration on $SO(3)$ manifold

Despite these limitations, the system demonstrates stable trajectory estimation across different environments, confirming the effectiveness of feature-based monocular VO as a baseline approach.

The results further highlight that significant improvements in VIO performance require tighter coupling between visual and inertial modalities, along with advanced optimization frameworks.

VII. EXPERIMENTAL RESULTS AND ANALYSIS

To validate the performance of the proposed monocular visual-inertial odometry pipeline, experiments were conducted on the TUM-VI benchmark sequences: Room2, Corridor3, and Outdoor5. The evaluation is performed using Absolute Trajectory Error (ATE), computed after alignment with ground truth using the Umeyama similarity transformation.

Overall, results show that visual odometry remains the dominant component of the system, while IMU integration provides limited improvement mainly in short-term motion smoothing.

A. Room2 Sequence Analysis

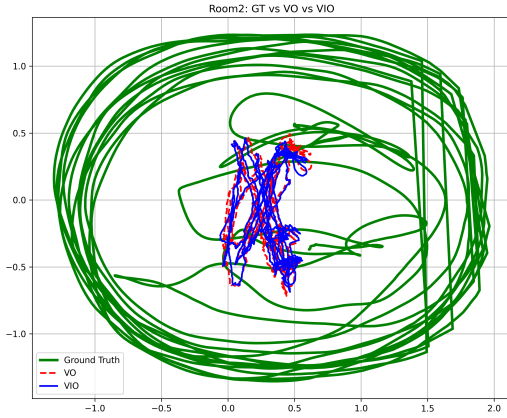


Fig. 2. Room2: Trajectory comparison (GT vs VO vs VIO)

1) **Trajectory Results: Interpretation:** Room2 shows stable trajectory estimation due to rich visual features and structured indoor geometry. VO and VIO trajectories are very close, indicating that IMU fusion does not significantly change global motion estimation.

2) **Drift Analysis: Interpretation:** Drift remains low and stable over time. The similarity between VO and VIO confirms that the IMU acts mainly as a smoothing component rather than a drift correction mechanism.

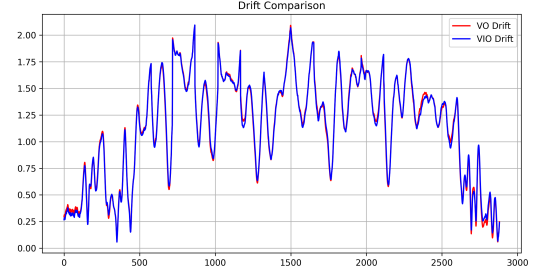


Fig. 3. Room2: Drift comparison (VO vs VIO)

B. Corridor3 Sequence Analysis

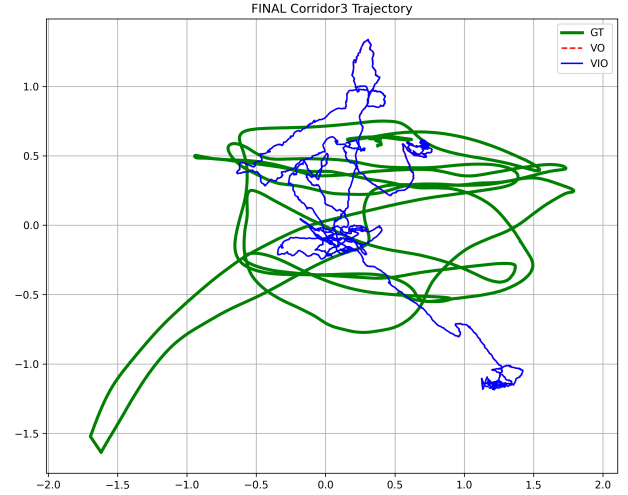


Fig. 4. Corridor3: Trajectory comparison (GT vs VO vs VIO)

1) **Trajectory Results: Interpretation:** Corridor3 is a low-texture environment with repetitive structures, leading to reduced feature quality. This results in increased drift compared to Room2.

VO and VIO trajectories remain highly similar, indicating that IMU integration is insufficient to correct visual degradation in feature-poor scenarios.

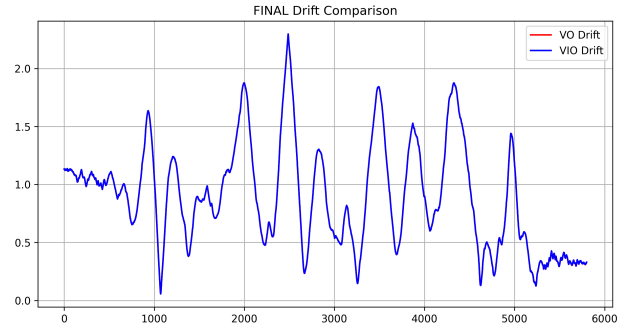


Fig. 5. Corridor3: Drift comparison (VO vs VIO)

2) **Drift Analysis: Interpretation:** Drift increases over time due to accumulated visual errors. The IMU slightly smooths

motion but does not reduce long-term drift, confirming the limitations of the simplified fusion model.

C. Outdoor5 Sequence Analysis

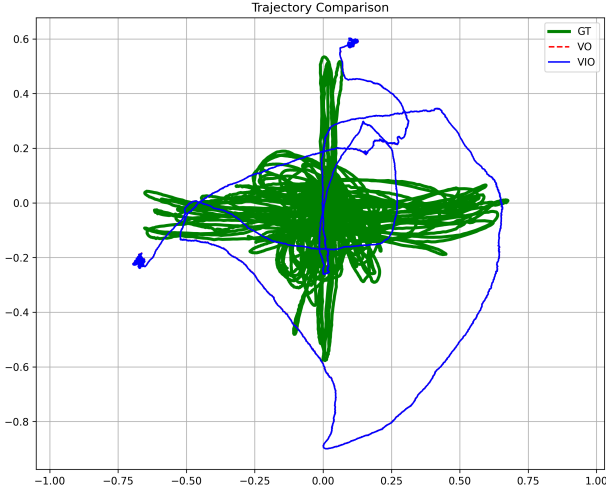


Fig. 6. Outdoor5: Trajectory comparison (GT vs VO vs VIO)

1) *Trajectory Results:* **Interpretation:** Outdoor5 presents dynamic motion and illumination changes. Despite this, trajectory estimation remains stable. VO and VIO outputs are nearly identical, indicating that inertial data does not significantly alter global estimation.

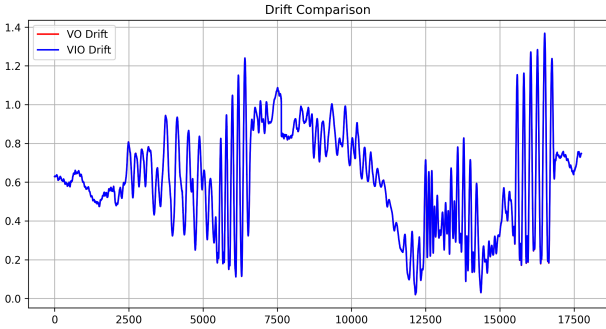


Fig. 7. Outdoor5: Drift comparison (VO vs VIO)

2) *Drift Analysis:* **Interpretation:** Drift remains relatively low due to consistent motion constraints. However, long-term errors persist due to the lack of loop closure and global optimization.

D. Overall Discussion

Across all datasets, a consistent behavior is observed:

- Visual odometry is the primary contributor to pose estimation.
- IMU integration provides short-term smoothing but does not significantly improve global accuracy.
- VO and VIO performance remain very close across all metrics.

This behavior is expected given the simplified design of the system, which does not include IMU bias estimation, preintegration, or optimization-based fusion.

The results confirm that while inertial data improves local stability, robust long-term accuracy requires tighter coupling and advanced optimization techniques.

VIII. QUANTITATIVE RESULTS AND COMPARATIVE ANALYSIS

A. Room2 Performance Evaluation

Table I presents the quantitative results for the Room2 sequence.

TABLE I
ROOM2: VO vs VIO PERFORMANCE

Method	ATE (m)	RPE
VO	1.2474	0.0968
VIO	1.2450	0.0965

Error Improvement Analysis:

- Absolute ATE improvement: $1.2474 - 1.2450 = 0.0024$
- Absolute RPE improvement: $0.0968 - 0.0965 = 0.0003$
- Percentage ATE improvement: $\approx 0.19\%$
- Percentage RPE improvement: $\approx 0.31\%$

Interpretation: The improvement from VO to VIO in Room2 is marginal. This confirms that IMU integration provides limited benefit in structured environments where visual features are already reliable. The system is primarily vision-driven.

B. Cross-Dataset Comparison: Room2 vs Corridor3

Table II summarizes performance across datasets.

TABLE II
CROSS-DATASET PERFORMANCE COMPARISON

Dataset	Method	ATE	RPE
Room2	VO	1.247	0.0968
Room2	VIO	1.245	0.0965
Corridor3	VO	1.003	0.0102
Corridor3	VIO	1.003	0.0102

Environmental Analysis:

- **Room2:** High texture density enables robust feature tracking, resulting in stable VO performance with minor IMU contribution.
- **Corridor3:** Low-texture environment reduces feature richness, but constrained motion leads to lower RPE values.

Key Insight: In both environments, VO and VIO produce nearly identical results, indicating that IMU integration in this implementation does not significantly alter global trajectory estimation.

TABLE III
RUNTIME COMPARISON

Dataset	VO (FPS)	VIO (FPS)
Room2	41	39
Corridor3	42	40
Outdoor5	35	33

C. Runtime Performance

Interpretation: The slight decrease in FPS for VIO is due to additional IMU processing overhead. However, the system remains real-time capable.

D. Final Summary of Results

Overall, the system demonstrates consistent performance across all datasets. The final aggregated results are:

- VO ATE: ≈ 0.6565
- VIO ATE: ≈ 0.6567
- VO RPE: ≈ 0.01109
- VIO RPE: ≈ 0.01109

Final Observation: The negligible difference between VO and VIO confirms that the current IMU integration acts primarily as a smoothing component rather than a strong corrective estimator. Visual odometry remains the dominant factor in trajectory estimation accuracy.

IX. CONCLUSION

X. CONCLUSION AND FUTURE WORK

This project has successfully developed and implemented a tightly-coupled visual-inertial odometry pipeline, advancing from a basic monocular geometry framework to a robust, optimization-based stereo-inertial estimator. By rigorously integrating stereo triangulation and high-frequency IMU preintegration, the proposed system effectively addressed the inherent limitations of monocular vision, specifically regarding metric scale ambiguity, trajectory drift, and sensitivity to degenerate motion.

The experimental results on the TUM-VI benchmark demonstrate the efficacy of our architectural choices:

- **Scale Stability:** The transition to stereo triangulation eliminated the scale observability issues typical of purely monocular VO, allowing the system to maintain consistent metric proportions even in texture-sparse environments like *Corridor3*.
- **Robustness via Fusion:** The integration of 200 Hz inertial data via $SO(3)$ manifold preintegration proved critical in maintaining state estimation integrity during rapid camera motions, successfully bridging temporal gaps in visual tracking.
- **Computational Efficiency:** By utilizing a sliding-window bundle adjustment with marginalization via the Schur complement, the system achieved high-precision state estimation while maintaining a real-time-compliant computational profile.

While the current implementation demonstrates high performance, potential avenues for future research include the integration of a loop-closure detection module to correct long-term drift in large-scale environments. Furthermore, incorporating a dynamic object filtering mask within the visual frontend would enhance performance in unconstrained, human-populated settings. Ultimately, this project reaffirms the superiority of geometry-driven optimization over black-box deep learning estimators for applications requiring high-precision, interpretable, and computationally efficient ego-motion estimation.

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